Causal Modeling in R: Whole Game

Malcolm Barrett
Stanford University

1. Specify causal question (e.g. target trial)
2. Draw assumptions (causal diagram)
3. Model assumptions (e.g., propensity)
4. Diagnose model (e.g., balance)
5. Estimate causal effects (e.g., IPW)
6. Sensitivity analysis

We’ll focus on the broader ideas behind each step and what they look like all together; we don’t expect you to fully digest each idea. We’ll spend the rest of the workshop taking up each step in detail

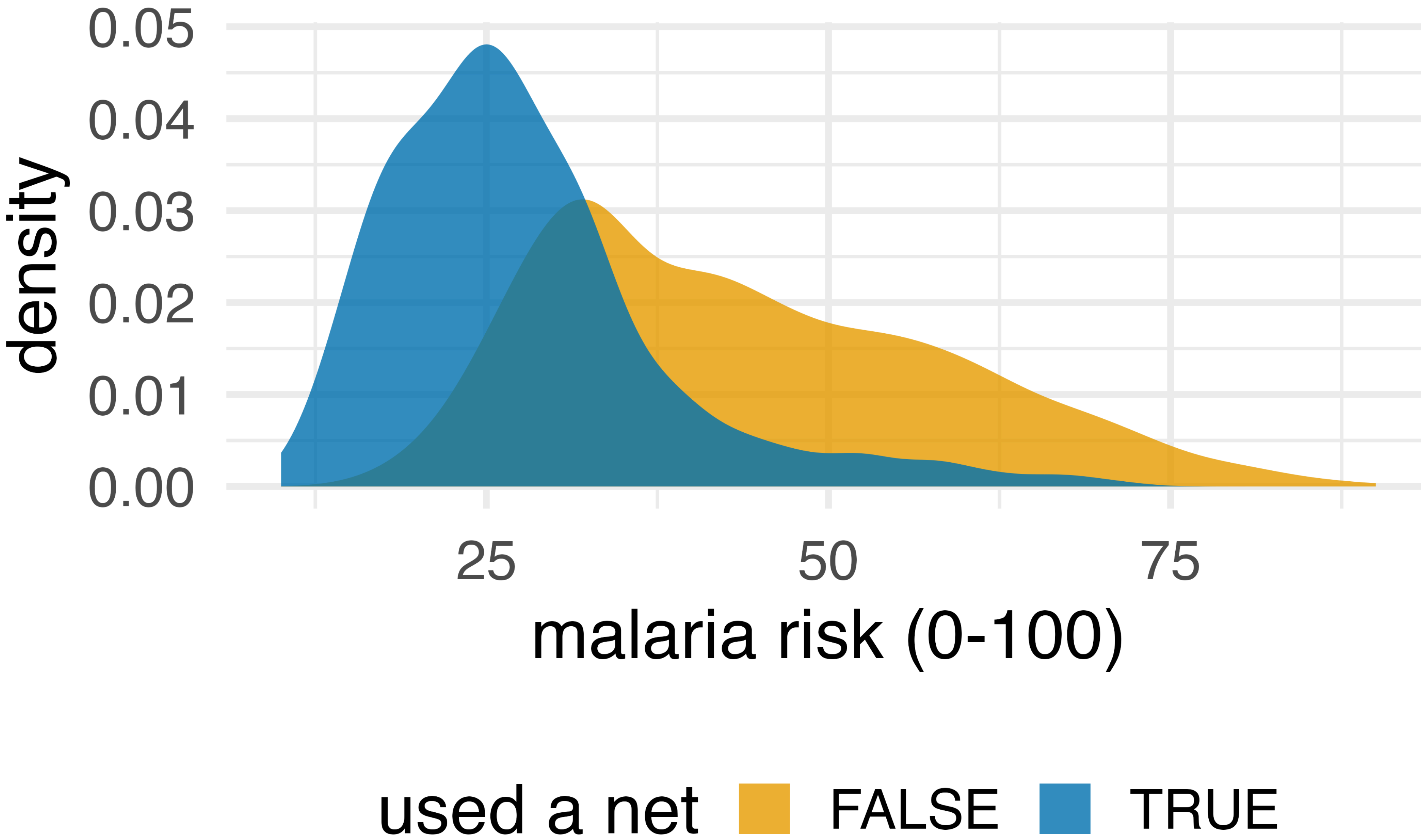
Does mosquito bed net use reduce malaria risk?

```r
library(causalworkshop)
net_data
```

```
# A tibble: 1,752 × 10
      id net   net_num malaria_risk income health household
   <int> <lgl>   <int>        <dbl>  <dbl>  <dbl>     <dbl>
 1     1 FALSE       0           38    779     35         1
 2     2 FALSE       0           48    700     35         3
 3     3 FALSE       0           32   1083     58         3
 4     4 FALSE       0           55    753     68         3
 5     5 FALSE       0           36    919     46         5
 6     6 FALSE       0           30    969     37         3
 7     7 FALSE       0           29   1012     58         1
 8     8 FALSE       0           45    708     30         2
 9     9 FALSE       0           51    733     18         3
10    10 FALSE       0           42    862     64         3
# ℹ 1,742 more rows
# ℹ 3 more variables: eligible <lgl>, temperature <dbl>,
#   insecticide_resistance <dbl>
```

Thanks to Andrew Heiss for the data!

Do net users have lower malaria risk?

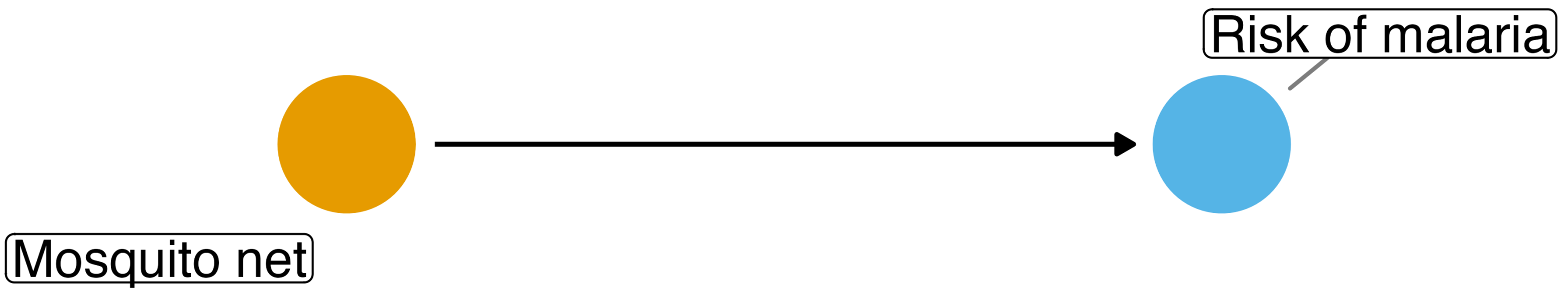


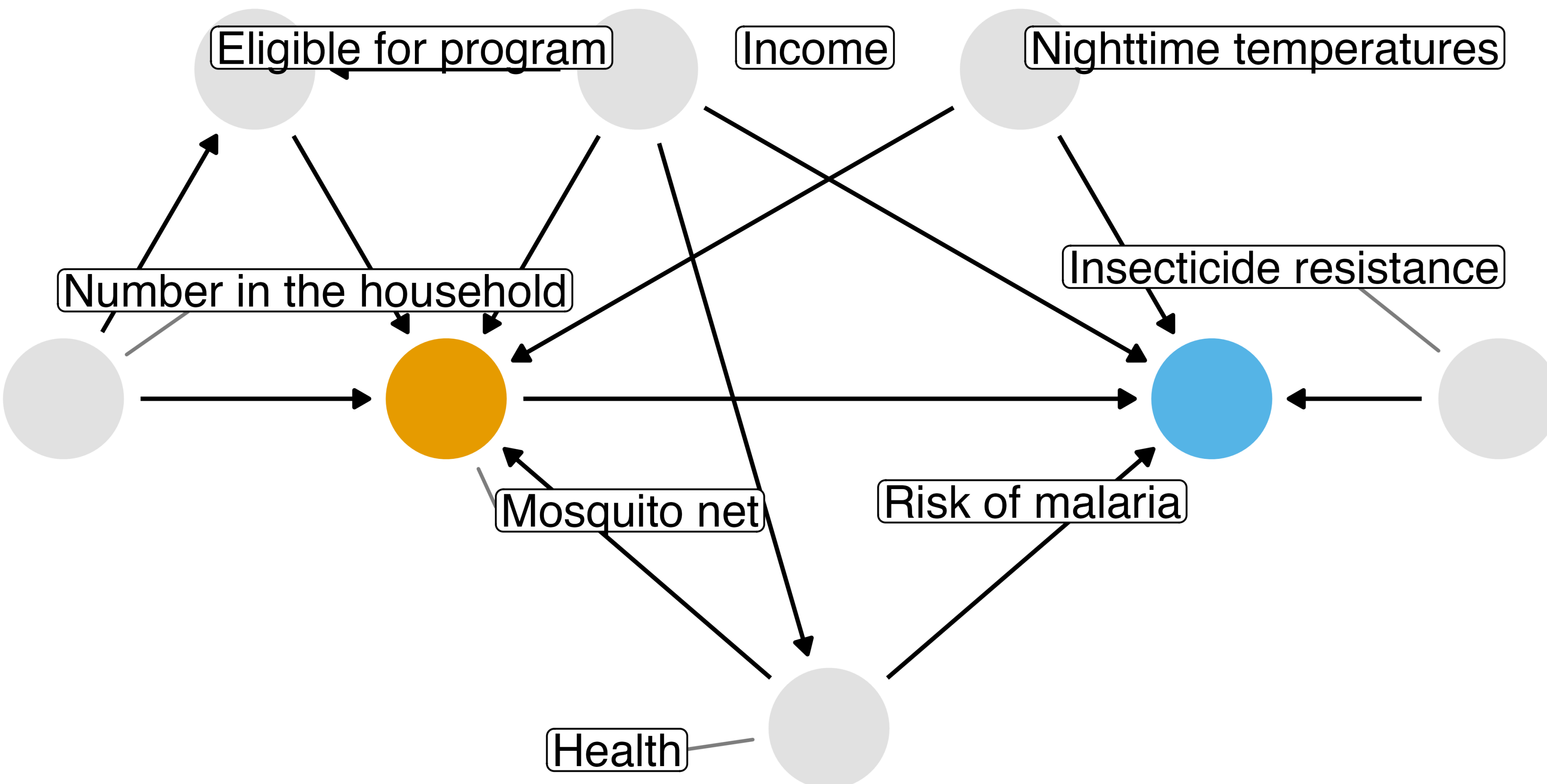
Do net users have lower malaria risk?

```r
# ~16 points lower risk for net users vs. non-users
net_data |>
  group_by(net) |>
  summarize(
    mean_malaria_risk = mean(malaria_risk),
    sd = sd(malaria_risk),
    .groups = "drop"
  )
```

```
# A tibble: 2 × 3
  net   mean_malaria_risk    sd
  <lgl>             <dbl> <dbl>
1 FALSE              43.9  14.6
2 TRUE               27.5  10.3
```

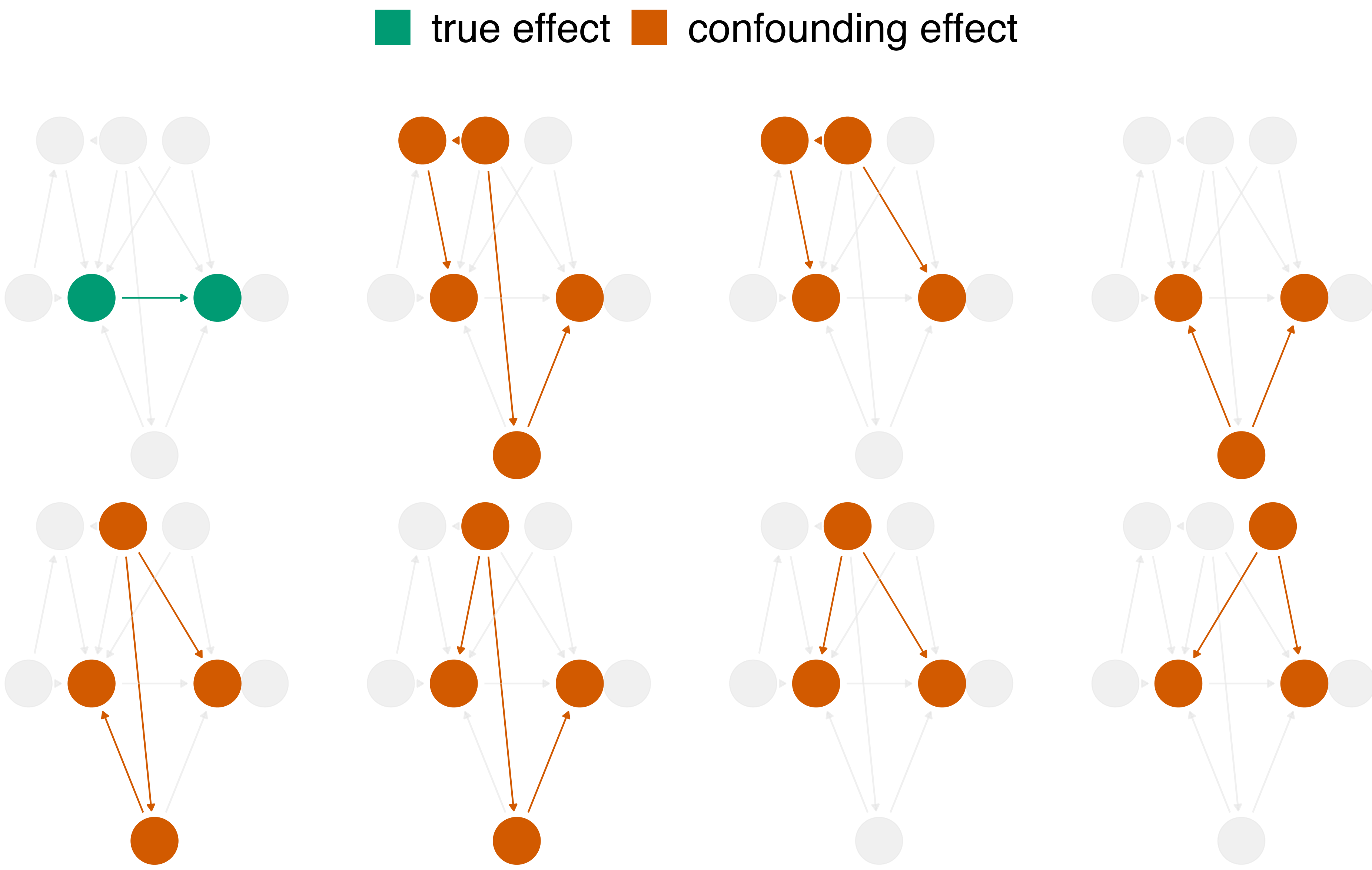
draw your assumptions

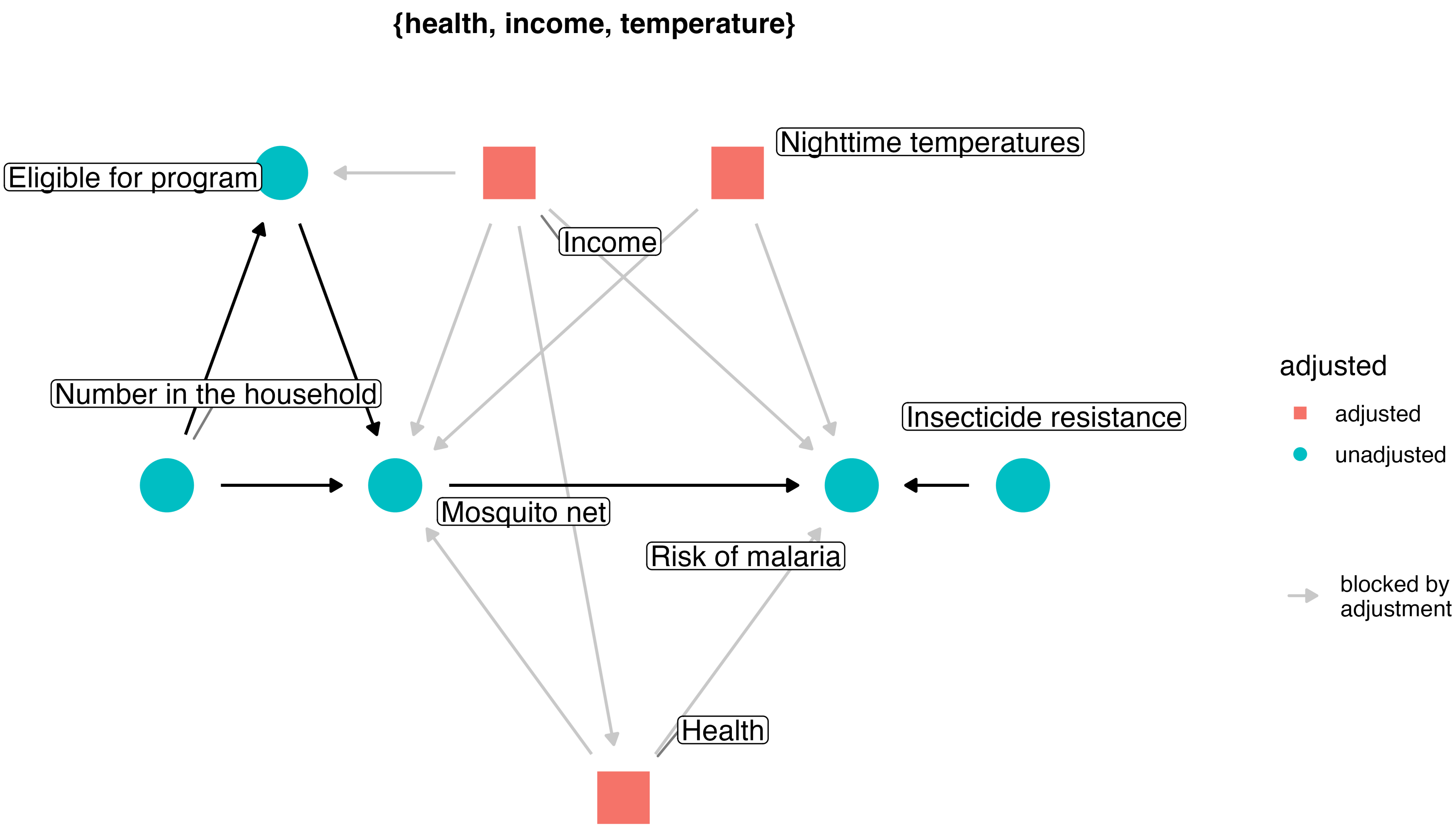

Risk of malaria
Mosquito net


Eligible for program
Income
Nighttime temperatures
Number in the household
Insecticide resistance
Mosquito net
Risk of malaria
Health

What do I need to control for?

One causal path, seven confounding





{health, income, temperature}
Eligible for program
Nighttime temperatures
Income
Number in the household
Insecticide resistance
Mosquito net
Risk of malaria
Health
adjusted
adjusted
unadjusted
blocked by adjustment

Multivariable regression: what's the association?

```
lm(
  malaria_risk ~ net + income + health + temperature,
  data = net_data
) |>
  tidy(conf.int = TRUE) |>
  filter(term == "netTRUE")
```

```
# A tibble: 1 × 7
  term    estimate std.error statistic   p.value conf.low
  <chr>      <dbl>     <dbl>     <dbl>     <dbl>    <dbl>
1 netTRUE    -12.0     0.314     -38.2 2.69e-232    -12.6
# ℹ 1 more variable: conf.high <dbl>
```

model your assumptions

counterfactual: what if <u>everyone</u> used a net vs. what if <u>no one</u> used a net

Fit propensity score model

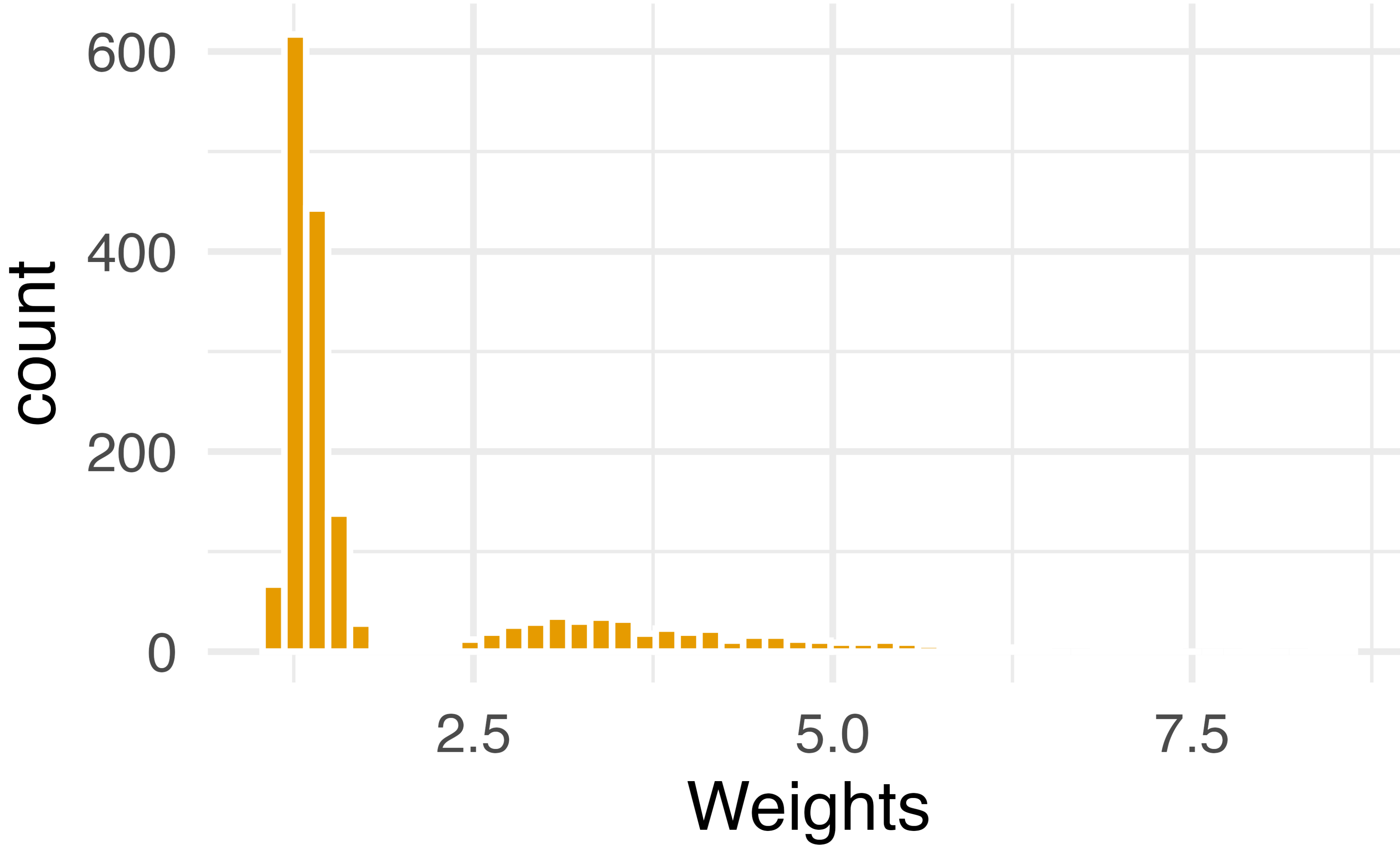
```r
propensity_model <- glm(
  net ~ income + health + temperature,
  family = binomial(),
  data = net_data
)
```

Calculate inverse probability weights

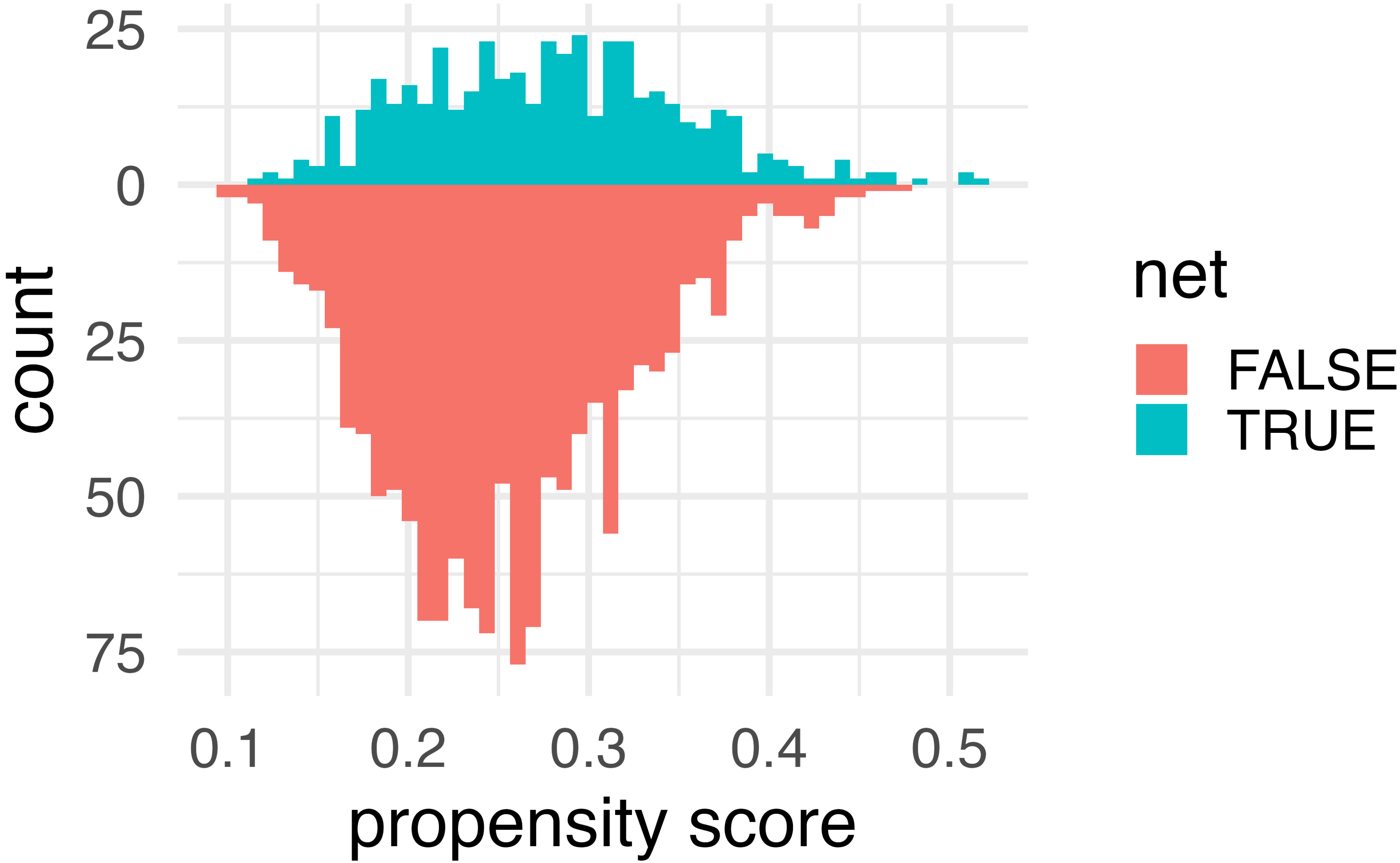
```r
library(propensity)
net_data_wts <- propensity_model |>
  # predict whether a household used a net
  augment(data = net_data, type.predict = "response") |>
  # calculate inverse probability
  mutate(wts = wt_ate(.fitted, net))
```

diagnose your model assumptions

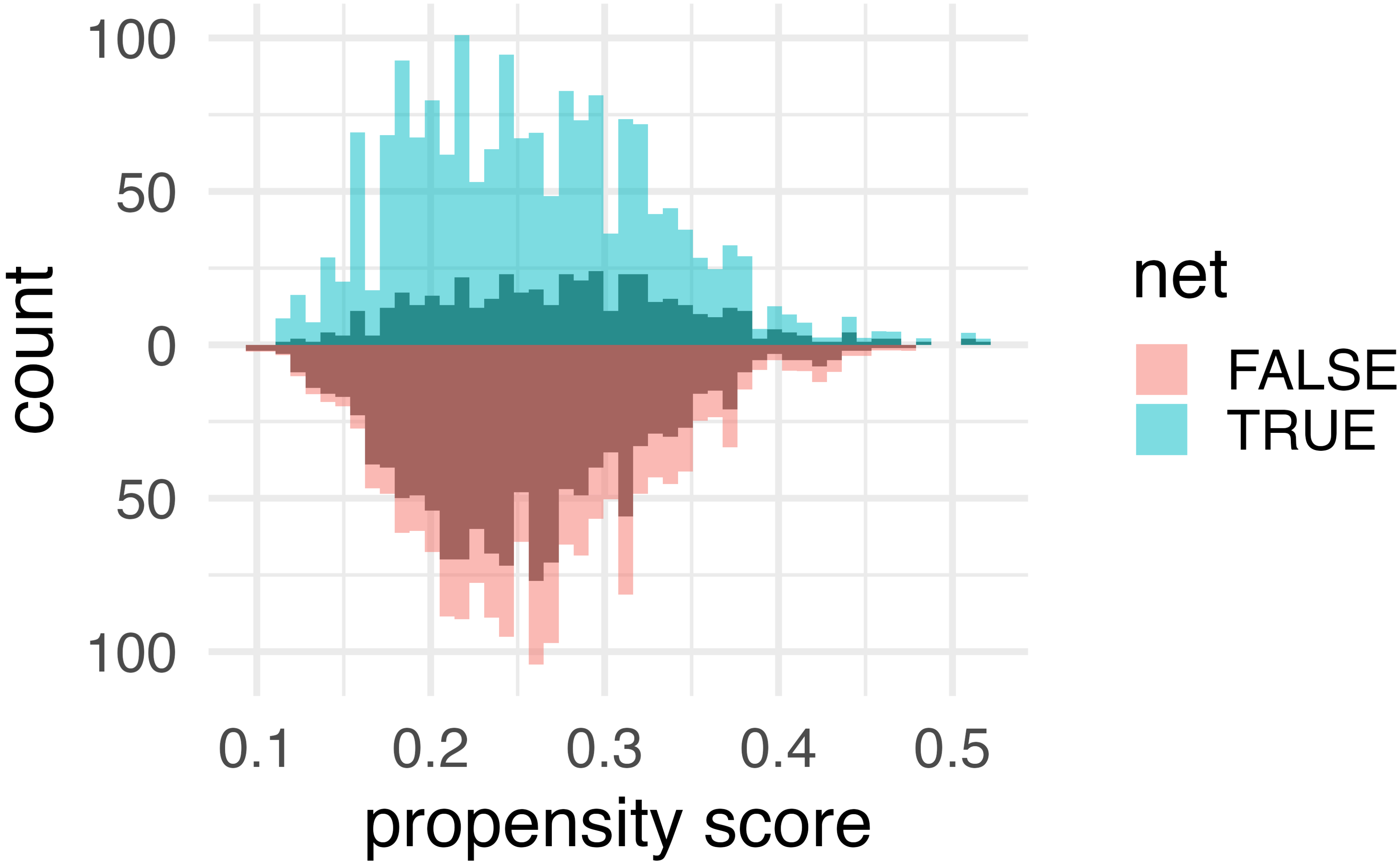
What's the distribution of weights?

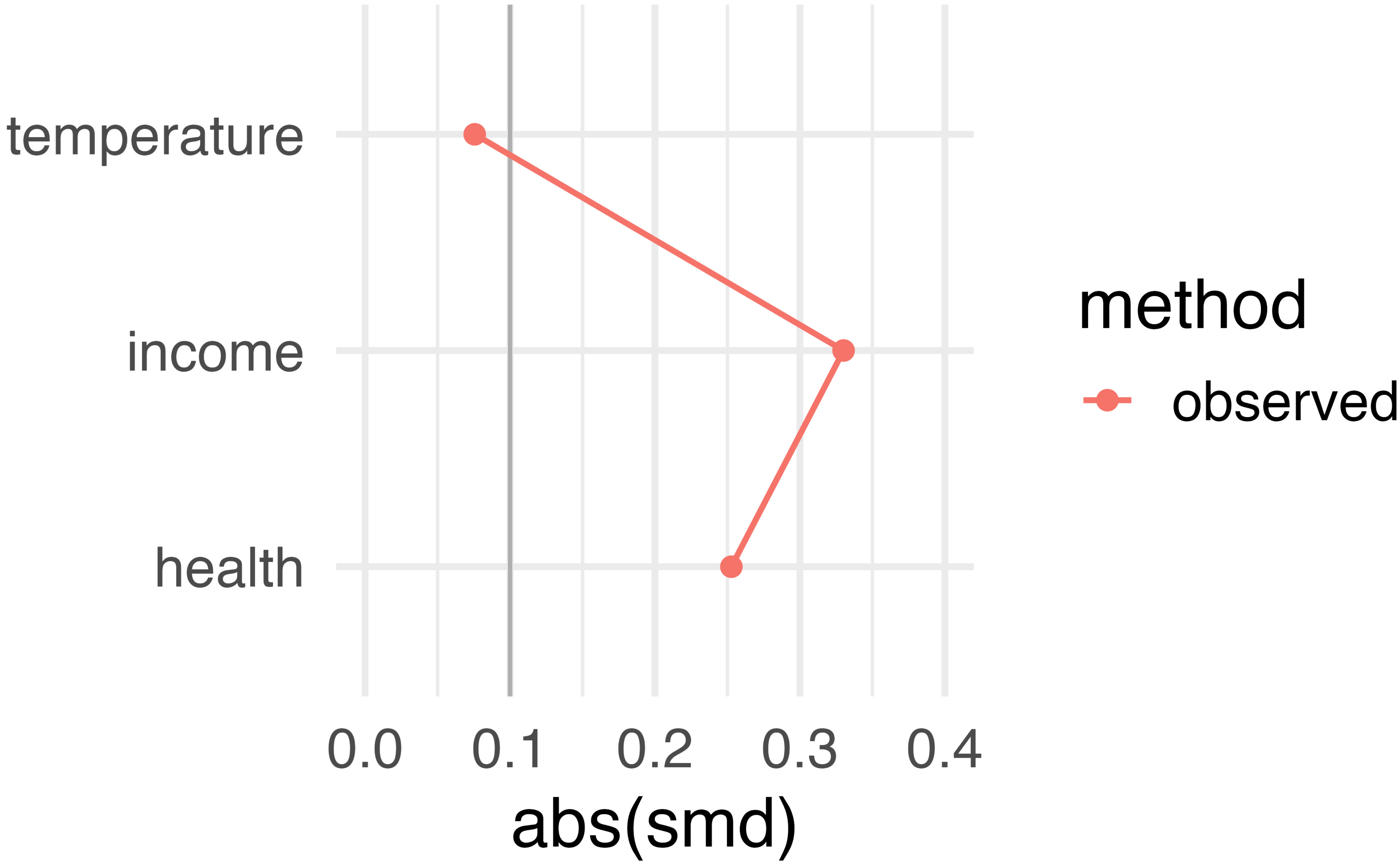


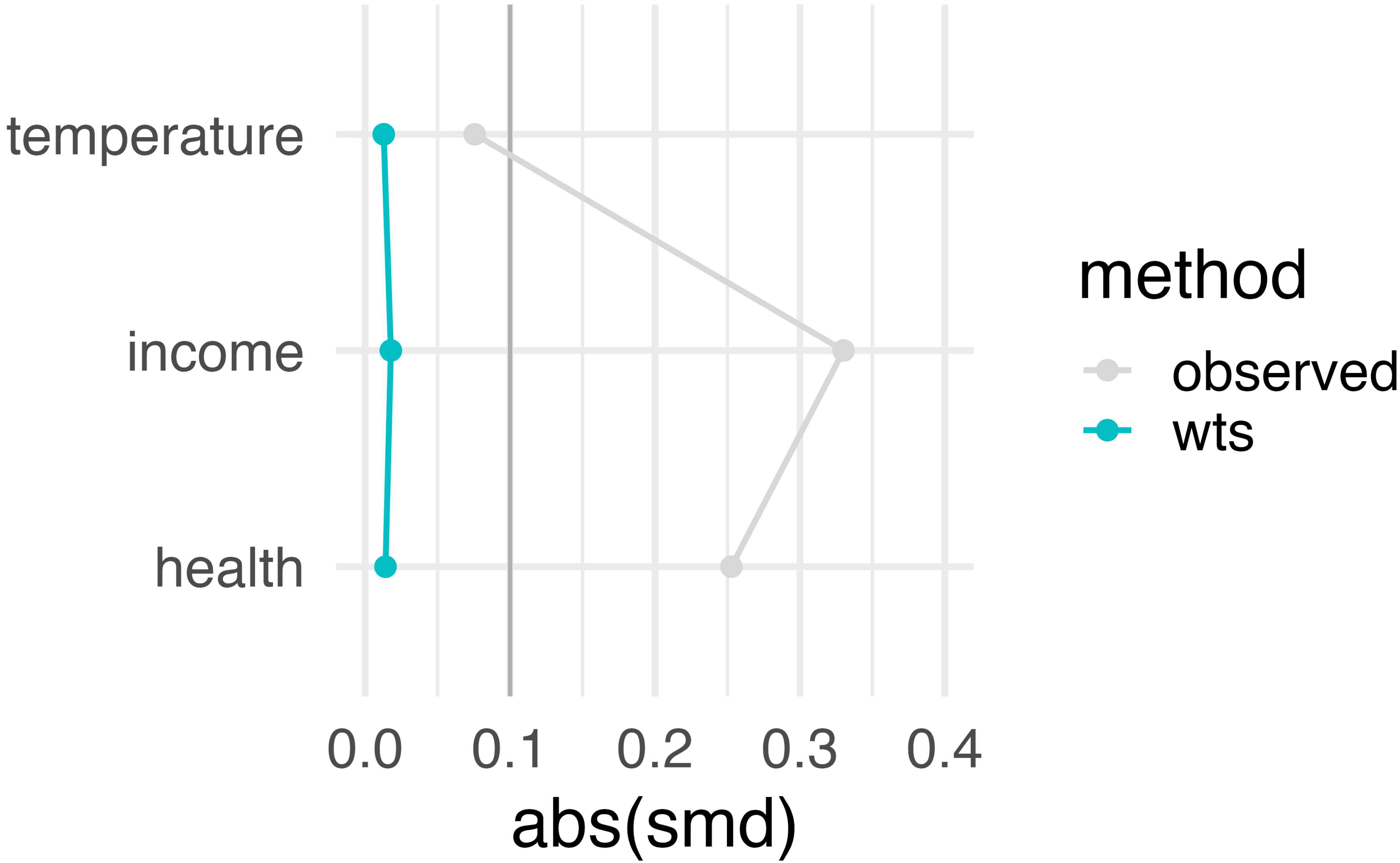
What are the weights doing?



What are the weights doing?




temperature
income
health
0.0
0.1
0.2
0.3
0.4
abs(smd)
method
observed


temperature
income
health
0.0
0.1
0.2
0.3
0.4
abs(smd)
method
observed
wts

estimate the causal effects

Estimate causal effect with IPW

```r
ipw_model <- lm(
  malaria_risk ~ net,
  data = net_data_wts,
  weights = wts
)

ipw_estimate <- ipw_model |>
  tidy(conf.int = TRUE) |>
  filter(term == "netTRUE")
```

Estimate causal effect with IPW

```
ipw_estimate
```

```
# A tibble: 1 × 7
  term    estimate std.error statistic  p.value conf.low
  <chr>      <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
1 netTRUE    -12.5     0.624     -20.1 5.50e-81    -13.8
# ℹ 1 more variable: conf.high <dbl>
```

Let’s fix our confidence intervals with robust SEs!

```r
# also see robustbase, survey, gee, and others
library(estimatr)
ipw_model_robust <- lm_robust(
  malaria_risk ~ net,
  data = net_data_wts,
  weights = wts
)

ipw_estimate_robust <- ipw_model_robust |>
  tidy(conf.int = TRUE) |>
  filter(term == "netTRUE")
```

Let’s fix our confidence intervals with robust SEs!

```
as_tibble(ipw_estimate_robust)
```

```
# A tibble: 1 × 9
  term    estimate std.error statistic  p.value conf.low
  <chr>      <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
1 netTRUE    -12.5     0.757     -16.6 2.36e-57    -14.0
# ℹ 3 more variables: conf.high <dbl>, df <dbl>,
#   outcome <chr>
```

Let's fix our confidence intervals with the bootstrap!

```r
# fit ipw model for a single bootstrap sample
fit_ipw_not_quite_rightly <- function(.split, ...) {
  # get bootstrapped data frame
  .df <- as.data.frame(.split)

  # fit ipw model
  lm(malaria_risk ~ net, data = .df, weights = wts) |>
    tidy()
}
```

```r
fit_ipw <- function(.split, ...) {
  # get bootstrapped data frame
  .df <- as.data.frame(.split)

  # fit propensity score model
  propensity_model <- glm(
    net ~ income + health + temperature,
    family = binomial(),
    data = .df
  )

  # calculate inverse probability weights
  .df <- propensity_model |>
    augment(type.predict = "response", data = .df) |>
    mutate(wts = wt_ate(.fitted, net))

  # fit correctly bootstrapped ipw model
  lm(malaria_risk ~ net, data = .df, weights = wts) |>
    tidy()
}
```

Using {rsample} to bootstrap our causal effect

```r
# fit ipw model to bootstrapped samples
ipw_results <- bootstraps(net_data, times = 1000, apparent = TRUE) |>
  mutate(results = map(splits, fit_ipw))
```

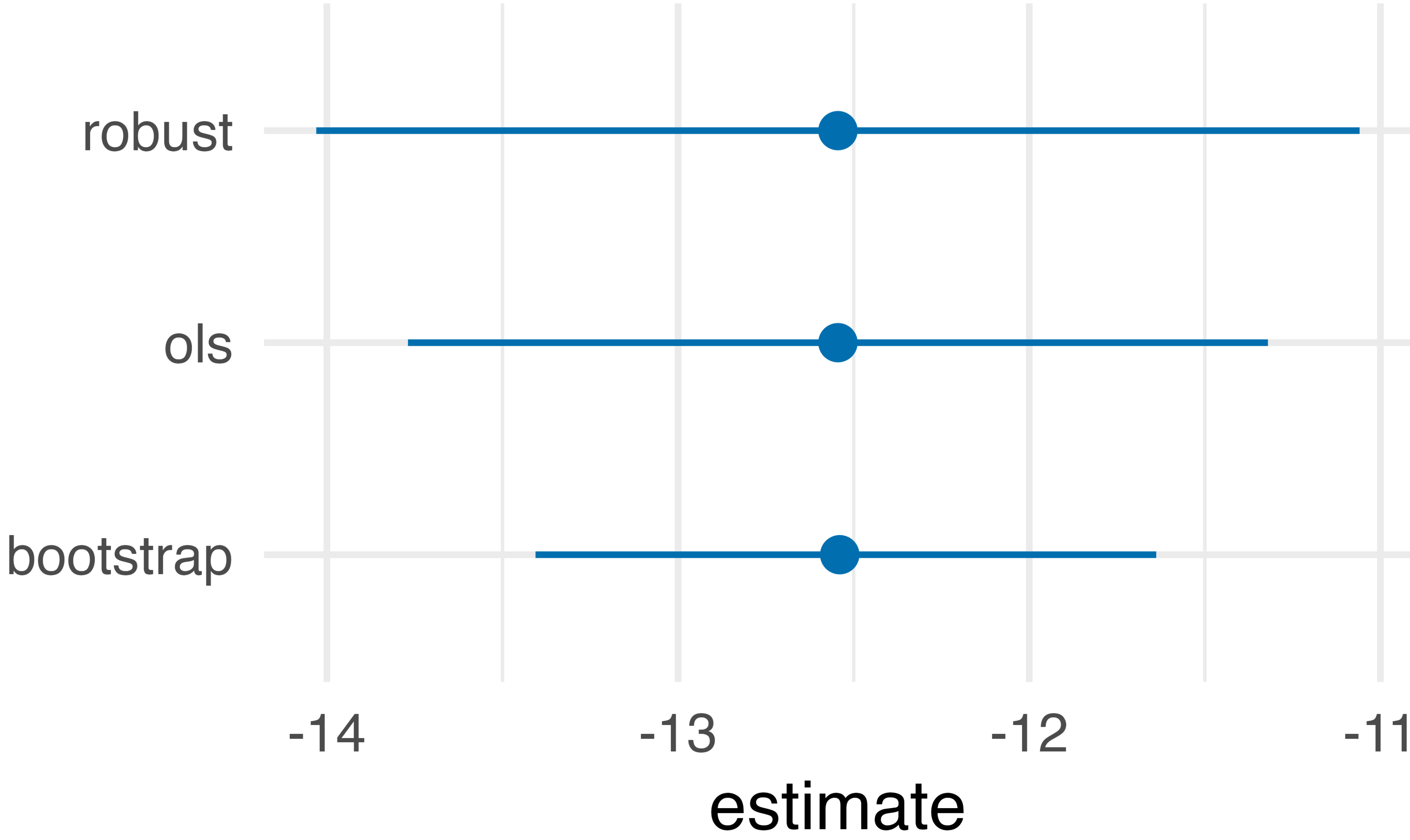
Using {rsample} to bootstrap our causal effect

```r
# get t-statistic-based CIs
boot_estimate <- int_t(ipw_results, results) |>
  filter(term == "netTRUE")

boot_estimate
```

Using {rsample} to bootstrap our causal effect

```
# A tibble: 1 × 6
  term    .lower .estimate .upper .alpha .method  
  <chr>    <dbl>     <dbl>  <dbl>  <dbl> <chr>    
1 netTRUE  -13.4     -12.5  -11.6   0.05 student-t
```


robust
ols
bootstrap
-14
-13
-12
-11
estimate

conduct a sensitivity analysis

Spoiler alert: the answer we just calculated is wrong

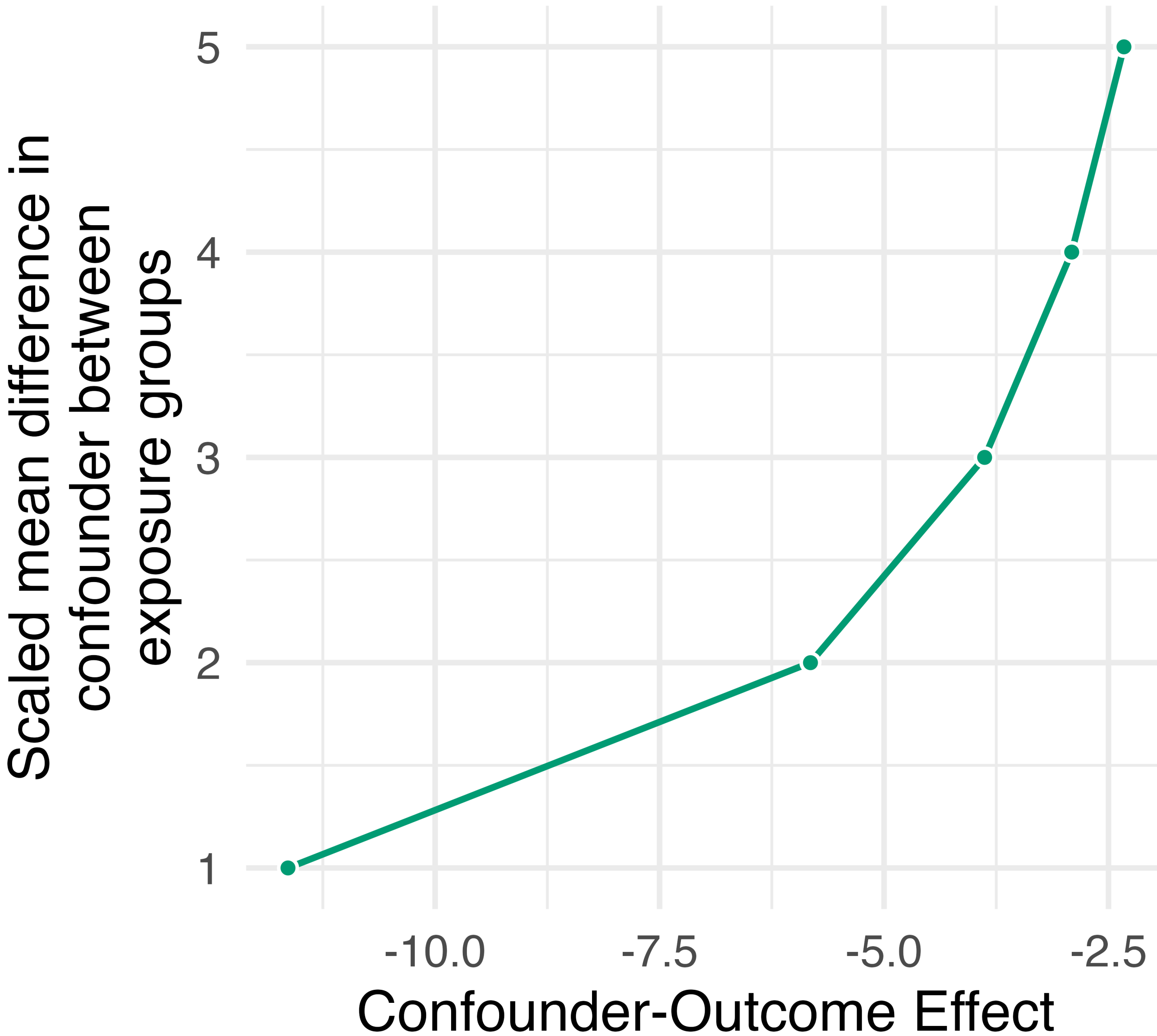
How much confounding would tip our estimate?

```r
library(tipr)
tipping_points <- tip_coef(
  boot_estimate$.upper,
  exposure_confounder_effect = 1:5
)

tipping_points |>
  select(exposure_confounder_effect, confounder_outcome_effect)
```

How much confounding would tip our estimate?

```
# A tibble: 5 × 2
  exposure_confounder_effect confounder_outcome_effect
                       <int>                     <dbl>
1                          1                    -11.6 
2                          2                     -5.82
3                          3                     -3.88
4                          4                     -2.91
5                          5                     -2.33
```


Scaled mean difference in confounder between exposure groups
5
4
3
2
1
-10.0
-7.5
-5.0
-2.5
Confounder-Outcome Effect

What would it take to explain away our estimate?

- A confounder with a scaled mean difference of 1 between net users and non-users would need an effect of -11.6 on malaria risk

- If the scaled mean difference were 5, an effect of -2.3 would be enough

An unmeasured confounder: genetic resistance to malaria

1. People with this genetic resistance have a malaria risk lower by about 10
2. About 26% of people who use nets have this genetic resistance
3. About 5% of people who don’t use nets have this genetic resistance

Adjust the estimate for the unmeasured confounder

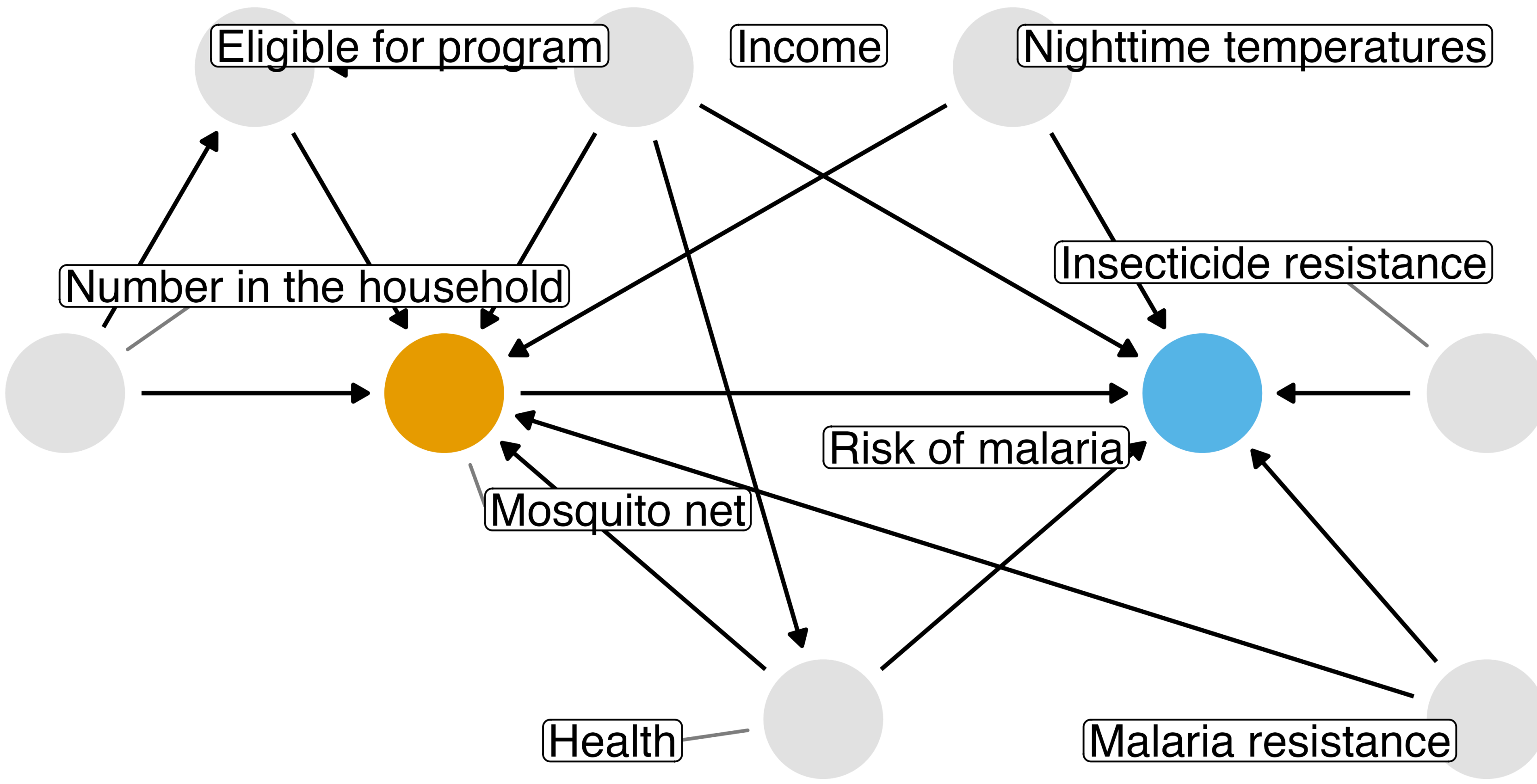
```
adjusted_estimates <- boot_estimate |>
  select(.estimate, .lower, .upper) |>
  unlist() |>
  adjust_coef_with_binary(
    exposed_confounder_prev = 0.26,
    unexposed_confounder_prev = 0.05,
    confounder_outcome_effect = -10
  )

adjusted_estimates |>
  select(effect_observed, effect_adjusted)
```

Adjust the estimate for the unmeasured confounder

```
# A tibble: 3 × 2
  effect_observed effect_adjusted
            <dbl>           <dbl>
1           -12.5          -10.4 
2           -13.4          -11.3 
3           -11.6           -9.54
```

The DAG that generated the data



```r
fit_ipw_full <- function(.split, ...) {
  # get bootstrapped data frame
  .df <- as.data.frame(.split)

  # add genetic resistance to the propensity score model
  propensity_model <- glm(
    net ~ income + health + temperature + genetic_resistance,
    family = binomial(),
    data = .df
  )

  # calculate inverse probability weights
  .df <- propensity_model |>
    augment(type.predict = "response", data = .df) |>
    mutate(wts = wt_ate(.fitted, net))

  # fit correctly bootstrapped ipw model
  lm(malaria_risk ~ net, data = .df, weights = wts) |>
    tidy()
}
```

Re-estimate the effect with the confounder measured

```r
ipw_results_full <- bootstraps(net_data_full, times = 1000, apparent = TRUE) |>
  mutate(results = map(splits, fit_ipw_full))

boot_estimate_full <- int_t(ipw_results_full, results) |>
  filter(term == "netTRUE")

boot_estimate_full
```

Re-estimate the effect with the confounder measured

```
# A tibble: 1 × 6
  term    .lower .estimate .upper .alpha .method
  <chr>    <dbl>     <dbl>  <dbl>  <dbl> <chr>
1 netTRUE  -11.1     -10.3  -9.31   0.05 student-t
```

These data are simulated. The true effect of net use on malaria risk is about -10

***Our causal effect estimate:* -10.3 (95% CI -11.1, -9.3)**

Review the Quarto file… later!

Resources

Causal Inference in R: **Our book, where this example comes from. Free online.**

Causal Inference: **Comprehensive text on causal inference. Free online.**

Bootstrap confidence intervals with {rsample}

R-causal: **Our GitHub org with R packages and examples**